Air

Exam code: 0620 & 0971

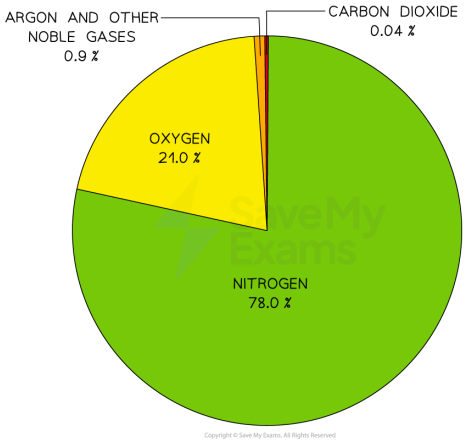
Download PDF

Written by: Caroline Carroll
Reviewed by: Stewart Hird
Updated on 17 January 2026

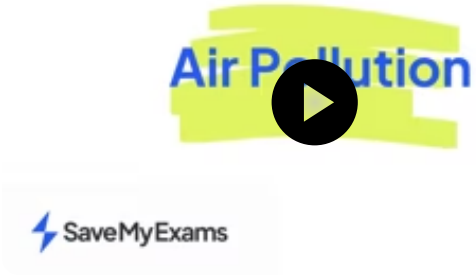
The composition of air

- The composition of gases in the atmosphere has not changed much in 200 million years
- The composition of clean, dry air is approximately:
  - 78% nitrogen
  - 21% oxygen
  - The remaining gases include carbon dioxide and trace quantities of the noble gases

Pie chart of the current atmosphere



The two main gases in the air are nitrogen and oxygen



Help me understand

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

Air pollution

- In addition to the gases present naturally in our atmosphere, other gases are present due to human activities and are classed as air pollutants

Carbon dioxide

- **Sources:**
  - Complete combustion of carbon-containing fuels such as fossil fuels
  - For example, the complete combustion of methane:

$$CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$$

- **Adverse effects:**
  - Increases global warming, which leads to climate change

Carbon monoxide

- **Sources**:
  - Incomplete combustion of carbon-containing fuels such as fossil fuels
  - For example, the incomplete combustion of gasoline / octane:

$$C_8H_{18} + 10O_2 \rightarrow 5CO + 3CO_2 + 9H_2O$$

- **Adverse effects**:
  - Toxic / poisonous
  - It combines with haemoglobin in the blood and prevents it from carrying oxygen

Particulates

- **Sources:**
  - Incomplete combustion of carbon-containing fuels such as fossil fuels can also produce particulates of carbon (soot)
  - For example, the incomplete combustion of methane can produce CO and C:

$$2CH_4 + 3O_2 \rightarrow 2CO + 4H_2O$$

$$CH_4 + O_2 \rightarrow C + 2H_2O$$

- **Adverse effects**:
  - Respiratory problems
  - Cancer

Methane

- **Sources:**
  - Waste gases from digestive processes of animals
  - Decomposition of vegetation
  - Bacterial action in swamps, rice paddy fields and landfill sites

- **Adverse effects**:

Oxides of nitrogen

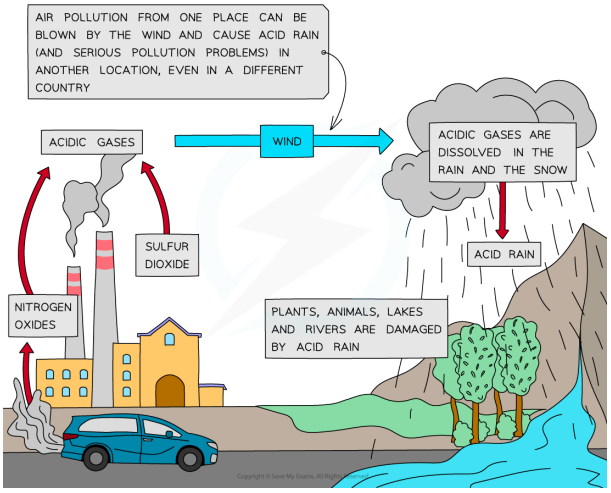
- **Sources**:
  - Reaction of nitrogen with oxygen in the presence of high temperatures, e.g. in car engines, high-temperature furnaces and when lightning occurs
  - It is also a product of bacterial action in the soil
- **Adverse effects**:
  - Produces photochemical smog
  - Dissolves in rain to form **acid rain** which causes corrosion to metal structures, buildings and statues made of carbonate rocks, damage to aquatic organisms
  - Pollutes crops and water supplies
  - Irritates lungs, throats and eyes and causes **respiratory** problems

Sulfur dioxide

- **Sources**:
  - Combustion of fossil fuels containing sulfur compounds:

$$S + O_2 \rightarrow SO_2$$

- Power stations are a major source of sulfur dioxide
- **Adverse effects**:
  - Dissolves in rain to form **acid rain** with similar effects as the acid rain caused by oxides of nitrogen



How acid rain is produced

Examiner Tips and Tricks

Complete and incomplete combustion of hydrocarbons produce different products.

- Complete combustion occurs in **excess** oxygen and produces CO_2 and H_2O.

Teacher only

New! Exclusive Teacher-Only Questions

- Fresh, expert-written questions
- Students can't access them
- Fair accurate tests - built fast

Explore in Test Builder

Test yourself

Flashcards

Was this revision note helpful? Yes No

Previous

Next

Course

Study tools

All topics

© Copyright 2015–2026 Save My Exams Ltd. All Rights Reserved.
Neither IBO nor College Board were involved in the production of, and do not endorse, the resources created by Save My Exams.